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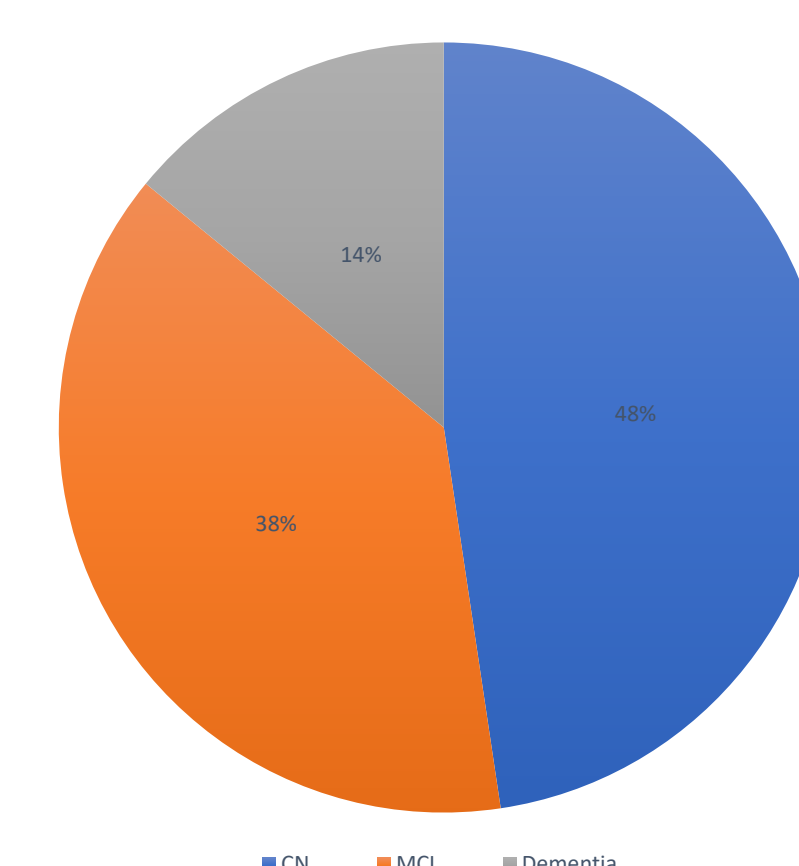
Introduction and Motivation

- Alzheimer's Disease (AD) is a progressive neurodegenerative disorder that leads to cognitive decline and brain connectivity disruptions due to brain tissue deterioration.
- This study investigates functional network connectivity across stages of cognitive decline—Cognitively Normal (CN, healthy individuals), Mild Cognitive Impairment (MCI, early memory/thinking issues), and Dementia (severe cognitive and functional decline)
- Identifying FNC features across these different cognitive decline stages can help understand disease progression.

Method

- FNC explores the mutual connections between different brain regions which are defined by simpler, time-related changes.
- ADNI dataset resting state fMRI scans of 2,462 subjects of CN, MCI and Dementia with 5,460 FNC features, were used which were used to examine alterations in neural connectivity associated with brain deterioration.
- Linear Mixed Effects (LME) modeling was chosen over Generalized Linear Models (GLM) as it accounts for subject-specific variability models FNC features while accounting age, sex, site and head motion.
- The study examines three group comparisons: Dementia vs. MCI, Dementia vs. CN, and MCI vs. CN, with Dementia as the reference group.

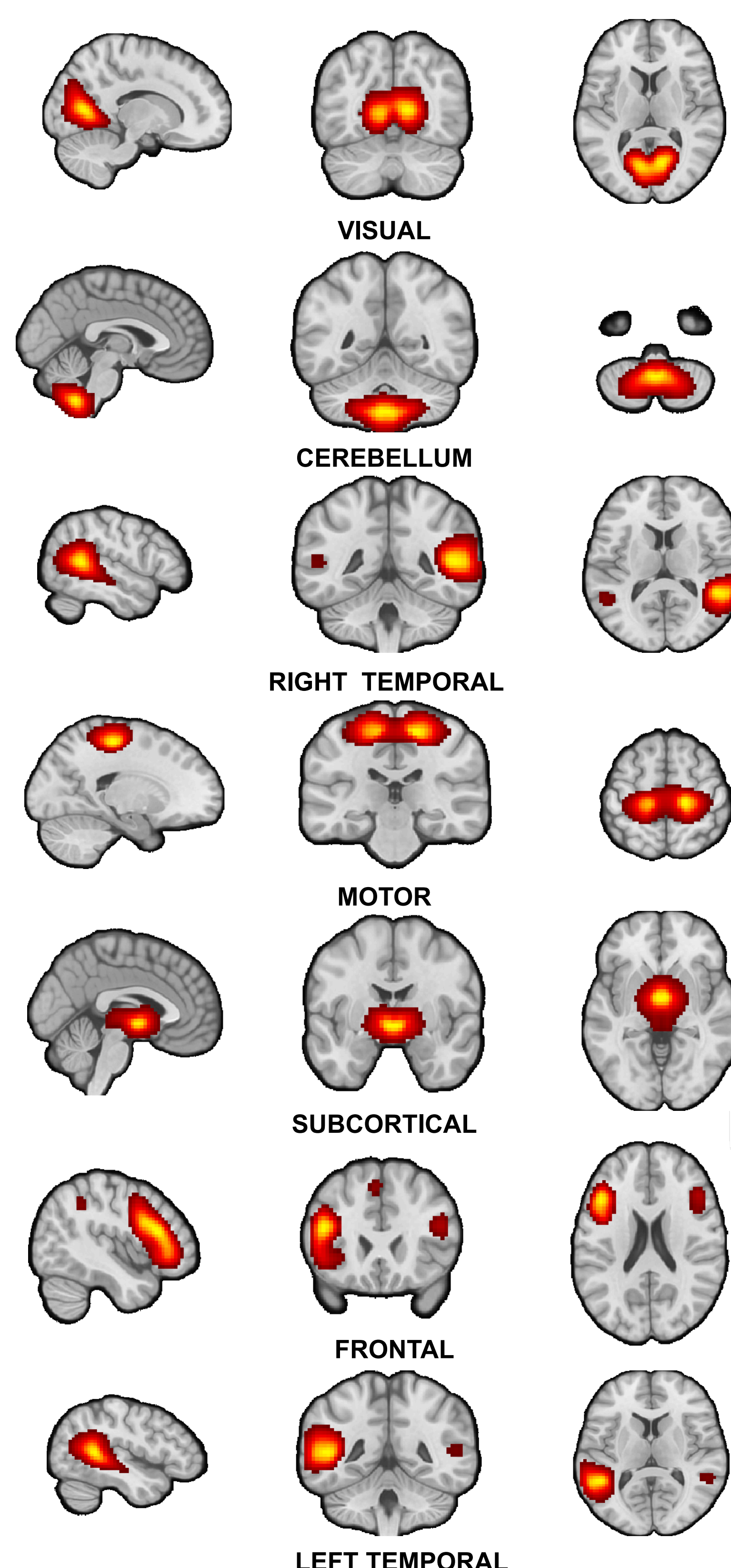
Percentage of Subjects per Disease Group



The figure above illustrates the percentage distribution of subjects across diagnostic groups: Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Dementia.

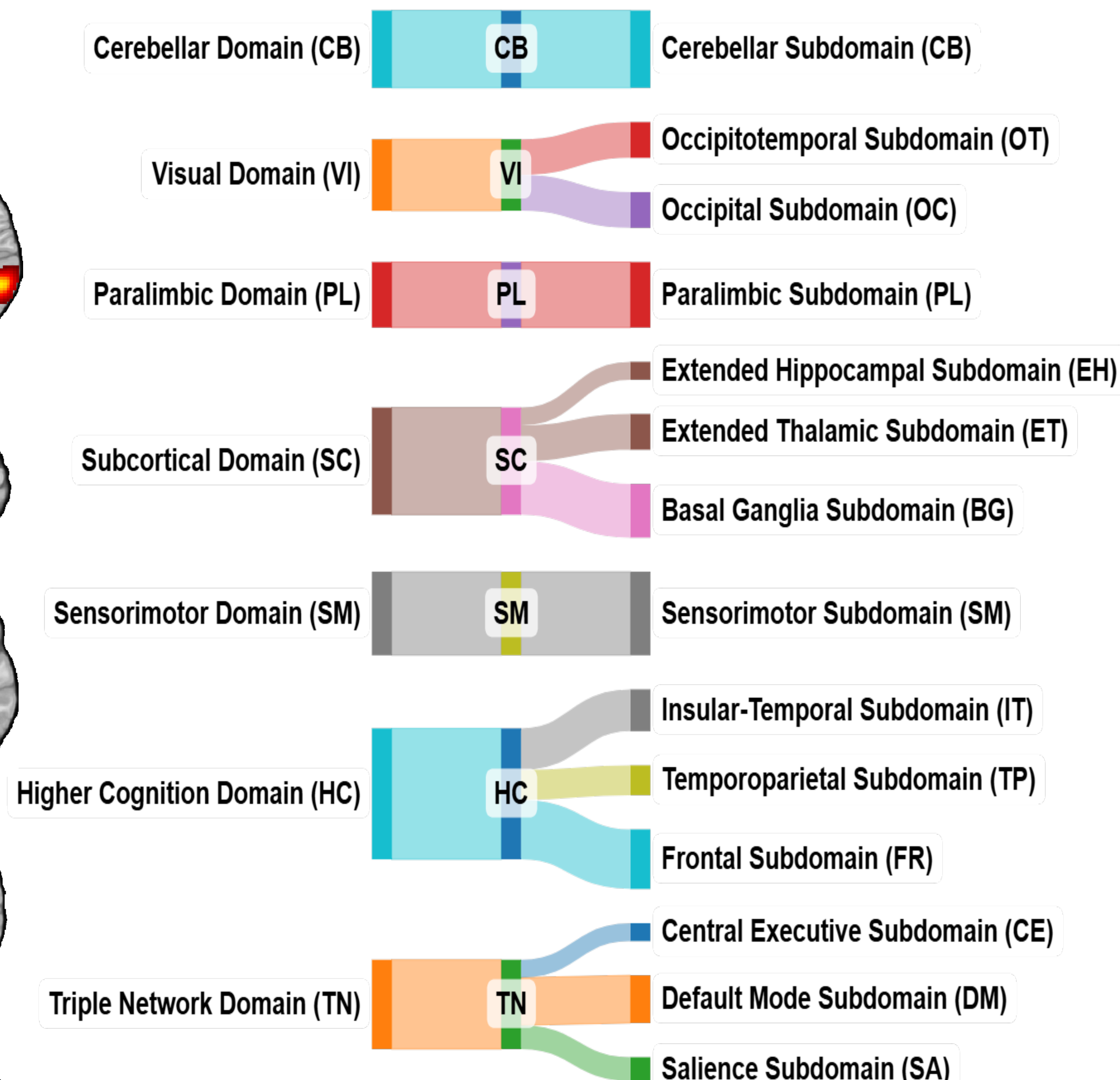
Results

SPATIAL MAPS FOR 7 COMPONENTS



The 7 identified components include the visual, cerebellum, left and right, temporal, motor, Frontal and Subcortical

Brain Domains Affected: Significant differences were observed across multiple brain domains.



FNC MATRICES

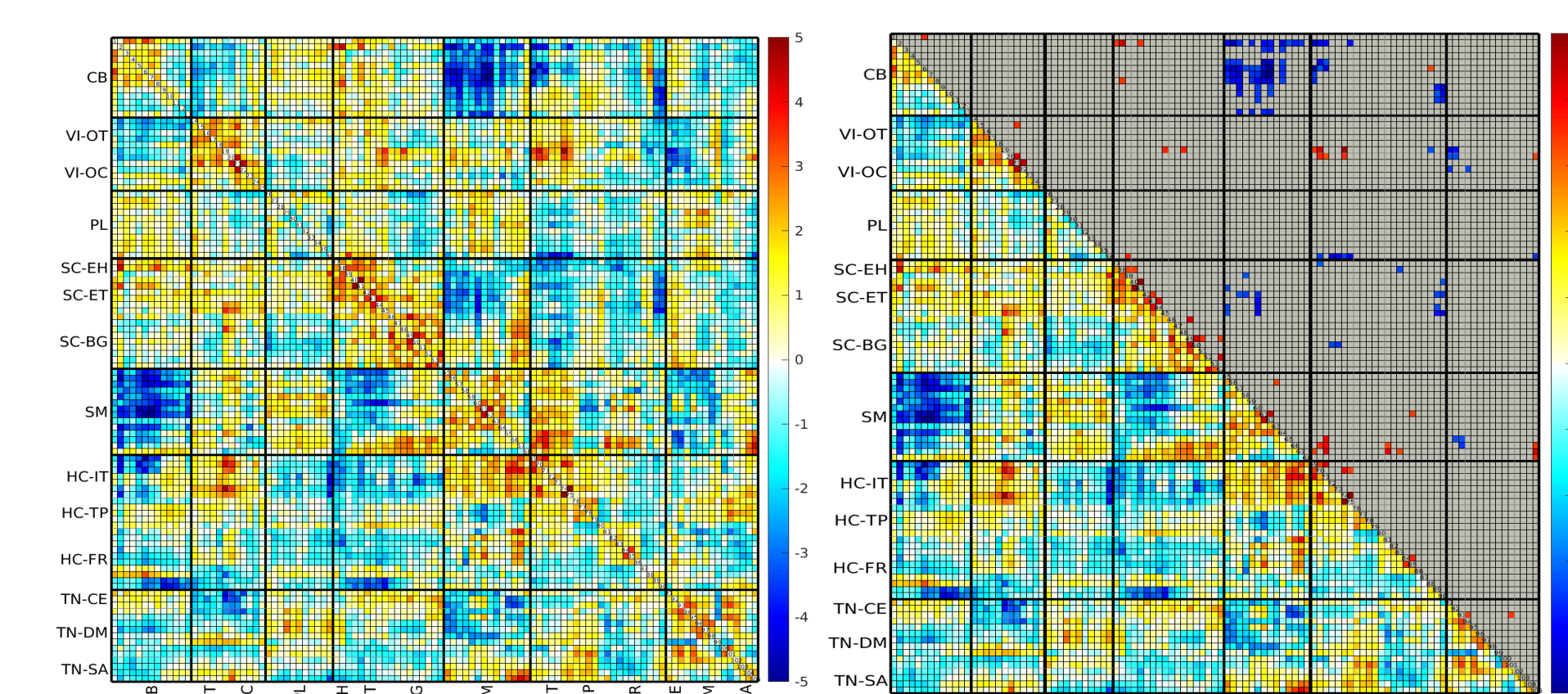


Figure 1
Dementia vs MCI

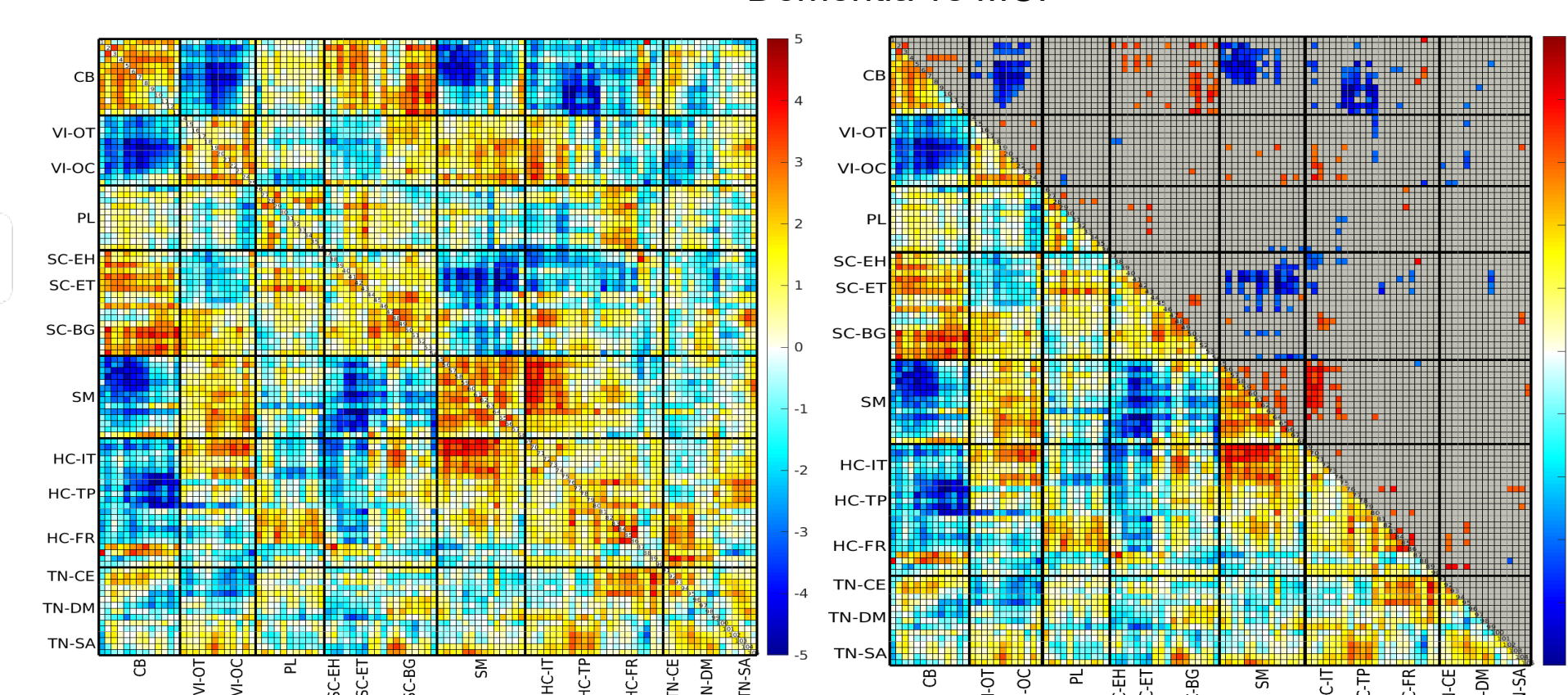


Figure 2
Dementia vs CN

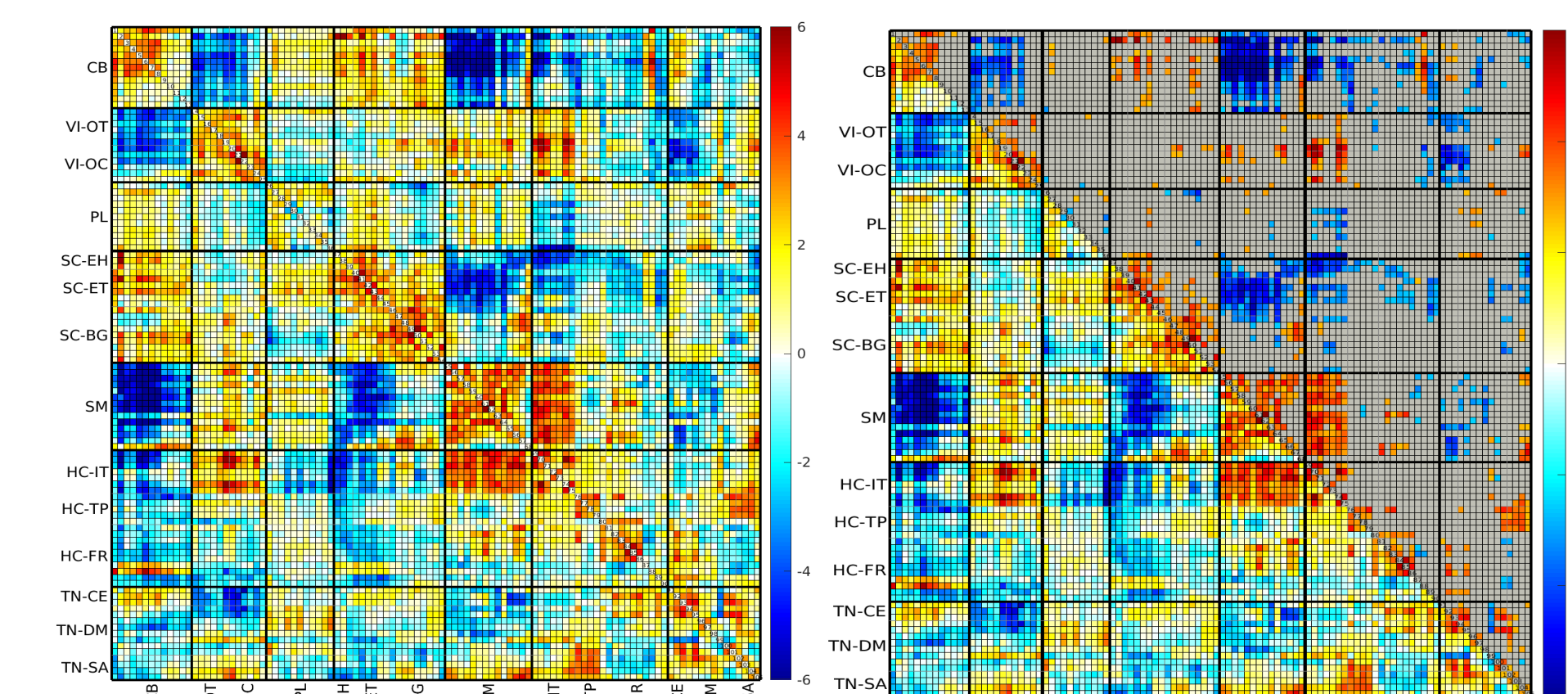


Figure 3
MCI vs CN

Figures 1, 2, and 3 illustrate connectivity trends among different groups using the Neuromark template. Figure 1 shows Dementia vs. MCI, Figure 2 shows Dementia vs. CN, and Figure 3 shows MCI vs. CN. In each figure, the left-hand side presents p-values, while the right-hand side highlights significant t-statistics, showcasing key differences in connectivity. Figure 2 demonstrates that the CN group exhibited stronger connectivity compared to the Dementia group. Figure 1 shows that the MCI group had stronger connectivity than the Dementia group. Figure 3 reveals that the CN group exhibited the strongest connectivity when compared to MCI.

Discussion

- The result findings indicate that FNC is significantly altered across different stages of cognitive impairment. We observe a progressive decline in disorder with the most significant FNC disruptions in the Dementia group.
- Weaker relationship between FNC of Dementia compared to MCI and CN suggests a progressive decline in network connectivity as cognitive impairment worsens,
- Overall, these results emphasize the potential of FNC as biomarker for tracking cognitive decline due to neurodegeneration and possibly identifying intervention targets

References

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